STAT 5200:

Applied Econometrics I

Fall 2026

Instructor:	Yisroel Cahn	**Section:**	001
Email:	ycahn@wharton.upenn.edu	**Dates:**	Aug. 25 - Dec. 7
Office:	Dinan 338	**Classroom:**	JMHH 245
Office Hours:	MW 3:30-4:30pm	**Time:**	MW 5:15-6:44pm

Overview: The aims of this course are to study basic econometric techniques. The emphasis will be upon the understanding and use of econometric methodology, and the written communication of the results of data analysis. Topics we will cover include conditional expectation, linear projection, potential outcomes, ordinary least squares estimation, instrumental variables estimation, systems of equations, panel data models, difference-in-differences methodology, and some issues in time series analysis. Some prediction and machine learning methods will be covered as well. We will explore mathematical and statistical foundations, as well as the application of statistical methodology. We will employ linear algebra extensively throughout, and will discuss and apply results from probability and theoretical statistics.

Course Material: The primary source for this course is the in class lectures. The required and optional readings will be placed in Canvas folders under the Readings folder.

- **Required:** *Econometric Analysis of Cross Section and Panel Data, 2nd ed.*, by Jefferey M. Wooldridge, MIT Press, 2010. This is the main course text. I expect to cover Chapters 1–6. You should be able to find a copy online.
- **(Optional:)** *Mostly Harmless Econometrics, An Empiricist's Companion*, by Joshua D. Angrist and Jorn-Steffen Pischke, Princeton University Press, 2009.
- **(Optional:)** *Causal Inference: The Mixtape*, by Scott Cunningham. Online version of the book is freely available from `https://mixtape.scunning.com/`
- **(Optional:)** *Econometrics*, by Fumio Hayashi, Princeton University Press, 2000.
- **(Optional:)** *Econometrics*, by Bruce Hansen, Princeton University Press, 2022.
- **(Optional:)** *The Elements of Statistical Learning: Data Mining, Inference, and Prediction, 2nd ed.*, by Trevor Hastie, Robert Tibshirani, and Jerome Friedman, Springer, 2009.

Software: The R package will be used in lectures and for homework. We will also use LaTeX for typesetting and Github for version control and collaboration, allowing students to develop reproducible and transparent research workflows that reflect current best practices in academia and industry.

Grading Policy:

Homework (50%), DataCamp Assignments (10%), Replication (20%), Project (20%)

Homework:

- There will be five (or, six) homework assignments. These will include theoretical exercises and involve the analysis of data and interpretation of the findings, and the presentation of well-organized and clearly written reports. **Homework must be submitted as PDFs typeset using LaTeX.** The homework is designed to teach and to give experience in the use of econometric methodology. You are encouraged to consult with each other in doing the homework, and also to contact me or the TA for help. **You must submit your own proofs, calculations, and your own writeup. Files should not be shared .** Homework must be submitted by the due date specified for the assignment. **All assignments will be submitted via Canvas.**

- Partial credit is given for problem sets turned in within 8 hours of the normal deadline in the following manner (determined by time-stamp for upload to Canvas): 20% reduction for any lateness, plus a reduction of 10% per hour, giving partial credit to any assignment turned in before 1:00AM the next day. (E.g., the score for an assignment turned in at 7:30pm would be reduced by 20%+25% = 45%.) Students are encouraged to turn in problem sets prior to the due date.

DataCamp Assignments:

- In addition to homework assignments, there will be DataCamp assignments. They will be assigned roughly weekly. The DataCamp invite link is: DataCamp.

- DataCamp assignments are graded based on completion.

Replication:

- This assignment will involve replicating the empirical analysis presented in a published research paper and presenting both the paper and replication results to the class. Students should select a paper that is closely related to the research project they plan to undertake at the end of the semester, allowing the replication exercise to serve as a foundation for their own work. Because students will be pursuing projects across a variety of fields, the chosen paper may come from any area of empirical research, provided it employs methods appropriate for the course. In addition to reproducing the main findings of the paper, students will summarize the research question, data, empirical strategy, and key results, and discuss the lessons learned from the replication process and how the paper informs their proposed project. In particular, students may find it useful to choose a paper that uses the same or a similar dataset, or that employs an empirical methodology they anticipate using in their own research.

- Students are encouraged to complete the replication assignment in groups of three to four and will deliver a 10-minute in-class presentation summarizing the paper, replication process, and findings. Students who prefer to complete the assignment individually may do so. However, because class time will not permit individual presentations from every student, those choosing the solo option will instead submit a recorded 10-minute presentation covering the same material.

- In addition, all students will complete three journal-style referee reports on papers selected by their classmates. These reports should emulate the peer review process used in academic publishing by providing a concise summary of the paper's research question and contribution, evaluating the strengths and weaknesses of the data, empirical strategy, and interpretation of the results, and offering constructive suggestions for improvement, extensions, or future research directions.

Project:

- Students will complete an original empirical research project using a dataset of their choosing and an empirical methodology appropriate for their research question. A key goal of this assignment is to develop the ability to organize, analyze, and communicate research findings through a clear and well-written report. Prior to beginning the project, students must submit a brief project proposal (no more than one page) outlining their research question, data, and proposed methodology for instructor approval. Additional details and project requirements will be provided later in the semester.

Email Policy:

- I try to respond to all emails within 48 hours (not including weekends or holidays).

Grading Scale and Distribution: The preliminary grade cutoffs based on total score are:

A+	97	B+	87	C+	77	D+	67	F:	<60
A	93	B	83	C	73	D	60		
A−	90	B−	80	C−	70				

Code of Academic Integrity: All students enrolled in courses in the Business Economics and Public Policy Department are expected to comply with the University of Pennsylvania's Code of Academic Integrity. We encourage all students to read the Code so that they are well aware of all situations that would be considered a violation. It is the policy of the Department of Business Economics and Public Policy to immediately fail any student who is to be in violation of the Code. Cheating, in any manner, on a graded assignment or exam will result in failing both the assignment/exam and the course. In addition to the sanctions imposed by the Department of Business Economics and Public Policy, the Office of Student Conduct may impose additional sanctions. Please review the Code of Academic Integrity as well as example of violations and possible sanctions: http://www.upenn.edu/provost/PennBook/academic_integrity_code_of

AI Policy: The use of artificial intelligence (AI) tools (e.g., ChatGPT, Claude, Gemini, Copilot, etc.) is permitted in this course as a learning aid. AI may be used to help generate ideas, explain concepts, improve writing, debug code, or provide feedback on your work.

However, AI should supplement (not substitute for) your own thinking and effort. All submitted work must reflect your understanding and analysis. Submitting AI-generated content as your own work, without meaningful engagement, revision, or verification, is not permitted. You are responsible for the accuracy, quality, and integrity of any work you submit, regardless of whether AI tools were used.

When in doubt, use AI as a tutor or assistant, not as a replacement for completing the assignment yourself.

Tentative Course Outline:

Tentative Outline (Subject to Change)			
Week	Class	Day	Topic
1	1	08/26/26	Introduction
2	2	08/31/26	R, LaTeX, and Github crash course
	3	09/02/26	Conditional expectations/linear projection
3		09/07/26	**No Class** *(Labor Day)*
	4	09/09/26	Conditional expectations/linear projection
4	5	09/14/26	Basic Asymptotic Theory
	6	09/16/26	Basic Asymptotic Theory
5	7	09/21/26	OLS estimation
	8	09/23/26	OLS estimation
6	9	09/28/26	Heteroskedasticity
	10	09/30/26	Bootstrapping
7	11	10/05/26	Omitted Variable Bias
	12	10/07/26	Measurement Error
8	13	10/12/26	Potential Outcomes Framework
	14	10/14/26	The Difference-in-Difference design (DiD)
9	15	10/19/26	The DiD design with staggered adoption
	16	10/21/26	Synthetic Controls
10	17	10/26/26	Regression Discontinuity Design
	18	10/28/26	Instrumental Variables
11	19	11/02/26	Instrumental Variables & 2SLS
	20	11/04/26	Introduction to Machine Learning

12	21	11/09/26	PAC Learning and VC Dimension
	22	11/11/26	**Replication Presentations**
13	23	11/16/26	**Replication Presentations**
	24	11/18/26	Logistic Regression
14		11/23/26	**No Class** *(Thnaksgiving Break)*
		11/25/26	**No Class** *(Thnaksgiving Break)*
15	25	11/30/26	Neural Networks and Deep Learning
	26	12/02/26	Decision Trees, Random Forests, and Gradient Boosting
16	27	12/07/26	Unsupervised Learning, Support Vector Machines, and Kernel Methods